PROJECT WATER

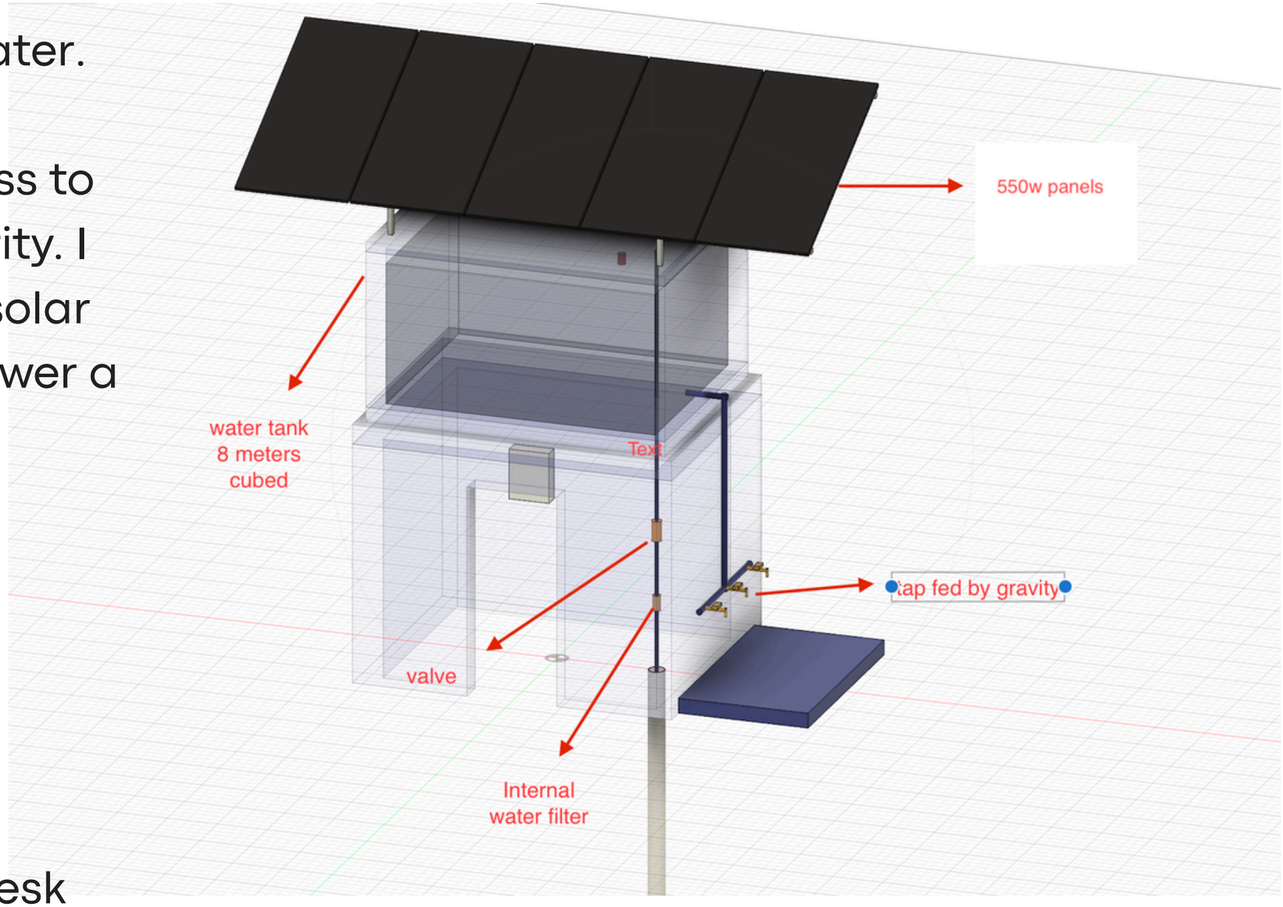
PROJECT WATER

Through my work with SolarPak, I began exploring how decentralized solar energy could address another challenge facing rural Sindh: unreliable access to water.

In the communities SolarPak serves, unreliable access to water remains a challenge alongside energy insecurity. I wanted to explore whether the same decentralised solar technology we used to electrify homes could also power a water system.

I wanted to develop a solution that could operate independently of an unreliable electricity grid while remaining practical for the communities we serve.

With no prior CAD experience, I taught myself Autodesk Fusion 360 in 2 weeks and developed an original 3D design for a solar-powered water pumping system.

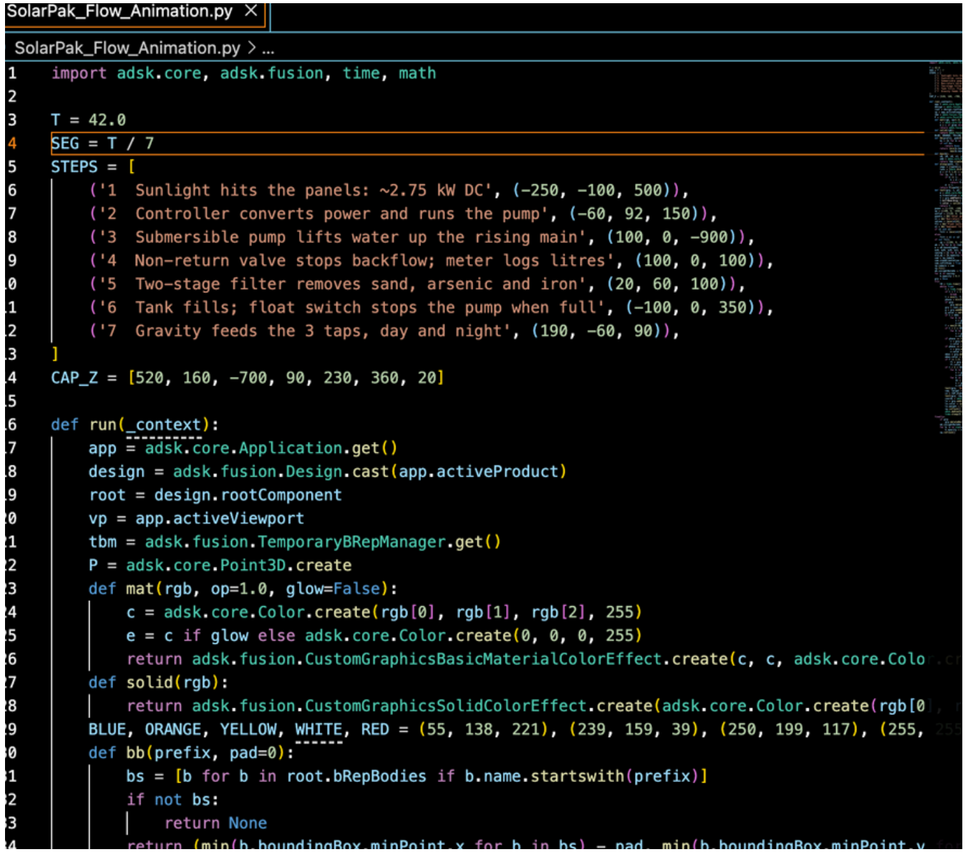


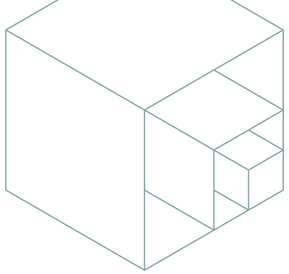
COMMUNICATING THE VISION

A technical design alone wasn't enough to secure organizational support.

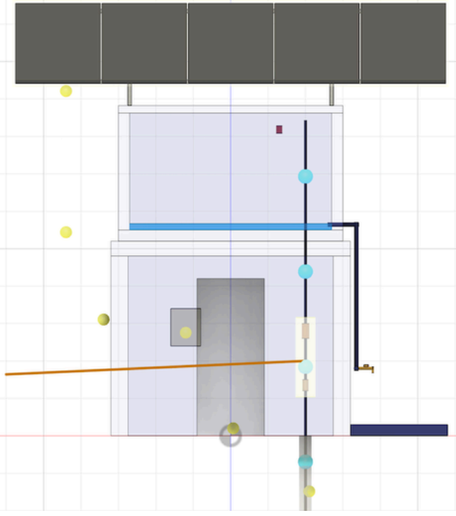
So I wrote a Python script using Fusion 360's API to animate each stage: solar generation, pumping, filtration, storage, and distribution.

I used it to walk SolarPak's six-member council through the system step by step.

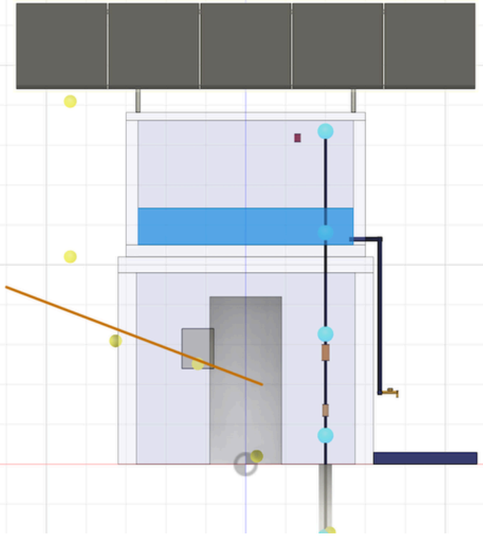




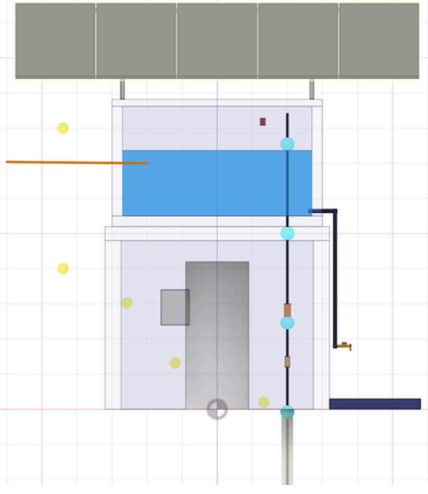
Frames from the animation ---->



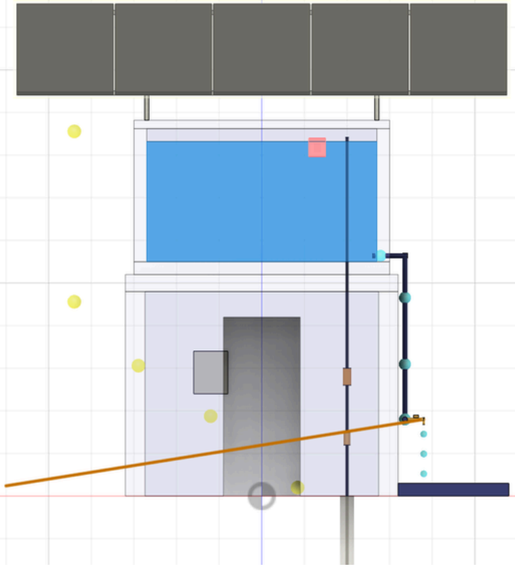
4 Non-return valve stops backflow; meter logs litres



5 Two-stage filter removes sand, arsenic and iron



6 Tank fills; float switch stops the pump when full



7 Gravity feeds the 3 taps, day and night

securing approval

With approximately PKR 400,000 raised through SolarPak's fundraising campaigns and recurring donor programmes, I proposed Project Water to our six-member decision-making council as an alternative to our regular household solar installations.
The council unanimously approved the project (6/6), taking it from an independently developed concept towards implementation.

We have identified a target village and aim to serve approximately 200 residents.
Implementation is tentatively planned for mid-November 2026, subject to technical verification and contractor arrangements.

What I am most proud of is not simply teaching myself CAD or developing the animation. It is taking something I initially explored independently and turning it into a proposal that six other people believed was worth pursuing. I learned that designing a solution and earning the support needed to implement it are two different challenges.